An Integrated AI and IoT-Based Framework for Automated Cement Quality Assessment

Sanchitha Lakshan
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
lakshansanchitha2000@gmail.com

Kanchana Koralage
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
itsmekanchanakoralge@gmail.com

Chamudini Abeywickrama
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
chamudiniayodya23@gmail.com

Hirumi Weerasooriya
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
hirumipraneethya@gmail.com

Dinuka Wijendra
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
dinuka.w@sliit.lk

Jenny Krishara
Faculty of Computing
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
jenny.k@sliit.lk

Roshantha Wickramasinghe
INSEE
Colombo, Sri Lanka
roshantha.wickramasinghe@siamcitycement.com

***Abstract*— The cement industry requires accurate and efficient quality control methods to ensure structural safety and material performance. This study proposes an integrated AI- and IoT-based framework for automated cement quality assessment. The system combines four main components: YOLOv9-based color particle identification for clinker microscopy analysis, IoT-based ultrasonic sensor measurement with U-Net-based crack detection for cement cube testing, multi-output cement strength prediction using an ensemble of XGBoost and LightGBM models, and automated clinker phase classification using MobileNetV2 with transfer learning. The system was validated using industrial datasets obtained from INSEE Cement (Private) Limited, Sri Lanka. Experimental results achieved 88% validation accuracy for particle identification, 90% validation accuracy for crack detection, 88% validation R^2 score for multi-age strength prediction, and 91% validation accuracy for clinker phase classification. The proposed framework enables faster and more reliable cement quality assessment suitable for industrial deployment.**

Keywords*— *cement quality control, clinker microscopy, crack detection, compressive strength prediction, industrial automation

I. Introduction

Cement is one of the most critical materials in modern construction, with global production exceeding 4 billion tons annually [1]. Quality control in cement manufacturing and concrete testing plays a vital role in ensuring structural safety, durability, and compliance with international standards such as ASTM, EN, and ISO specifications [2]. Even minor variations in chemical composition, fineness, or curing conditions can significantly affect the mechanical performance and long-term behavior of cement-based structures [3].

Despite its importance, traditional cement quality assessment methods face significant challenges related to efficiency, accuracy, and reliability. Microscopic examination of cement phases and clinker particles remains a fundamental practice in quality control; however, it is difficult to standardize across operators and requires a high level of expertise. Manual analysis involves visual inspection using optical microscopes, subjective interpretation of particle colors and phase boundaries, and time-consuming calculations, all of which are prone to human error and inconsistency [4]. Furthermore, these procedures are labor-intensive and unsuitable for high-throughput industrial environments.

In addition to microscopic analysis, compressive strength testing of cement and concrete cubes remains a time-dependent process, typically requiring 7-day and 28-day curing periods before reliable results can be obtained. This delay restricts early decision-making in production control and increases the risk of material wastage and process inefficiencies [5]. Early-age strength prediction and automated crack detection have therefore gained increasing attention as effective solutions for reducing testing time while maintaining accuracy [6].

Recent advancements in artificial intelligence (AI), machine learning (ML), computer vision, and Internet of Things (IoT) technologies have opened new opportunities for automating cement quality assessment and structural evaluation [7], [8], [10], [12], [13]. Deep learning-based image analysis has demonstrated high accuracy in material phase classification and defect detection, while IoT-enabled sensors provide real-time, non-contact measurements for physical testing applications [7], [8]. However, most existing studies address these challenges in isolation, focusing on individual tasks such as strength prediction, crack detection, or phase classification, without offering an integrated, end-to-end solution suitable for industrial deployment.

This research presents an integrated intelligent cement quality control system that combines four critical components, namely: automated color-based particle identification in clinker microscopy images for detecting red (calcite–$CaCO_3$) and white (quartz–SiO_2) particles; IoT-based non-contact area measurement with real-time crack detection for cement cube compressive strength testing; machine learning-based cement strength prediction incorporating environmental sensitivity; and automated classification of cement clinker phases (C_3S, C_2S, C_3A, and C_4AF) using deep learning techniques. By integrating these capabilities into a comprehensive web-based platform, the proposed system provides construction laboratories, cement manufacturers, and researchers with a scalable, accurate, and efficient solution for cement quality.

The problem addressed in this study arises from the inefficiencies of conventional cement quality control processes, which are largely manual, fragmented, and subjective. These limitations lead to inconsistencies, delayed feedback, and reduced responsiveness in industrial quality assurance workflows [9]. The proposed system addresses these challenges by automating and integrating the entire workflow using AI- and IoT-driven techniques, thereby enhancing accuracy, reducing human intervention, and enabling timely, data-driven quality control decisions.

II. Literature Review

The quality of cement production is highly dependent on the composition, distribution, and microstructural characteristics of particles in cement raw meal and clinker materials. Traditionally, microscopic analysis of cement raw meal and clinker phases has been performed manually by laboratory experts. Although effective, this process is time-consuming, subjective, and prone to human error, leading to inconsistencies in quality evaluation across operators and laboratories [10], [18]. With recent advancements in machine learning and computer vision, automated image analysis has emerged as a reliable alternative for industrial quality control. Deep learning–based object detection and classification models have demonstrated strong performance in identifying complex visual patterns in microscopic images, enabling more consistent and efficient particle and phase identification compared to manual inspection methods [11], [19].

Compressive strength is one of the most critical performance indicators in cement and concrete applications, directly influencing durability, safety, and structural service life. Conventional laboratory-based testing methods typically provide strength results only after predefined curing periods, limiting early-stage quality control and production optimization [12]. To address this limitation, recent studies have explored the use of machine learning and computer vision techniques for surface analysis and crack detection in cement and concrete specimens. Digital image processing combined with artificial neural networks has demonstrated high accuracy in estimating compressive strength from surface images [13]. In addition, deep learning architectures such as InceptionV3 have shown strong performance in detecting cracks in laboratory-scale concrete cubes, achieving accuracies exceeding 97% [14]. More recent research has demonstrated the effectiveness of U-Net, hybrid CNN architectures, and advanced segmentation techniques for fine-grained crack detection, damage quantification, and structural health monitoring in concrete materials [20], [21], [23].

Machine learning techniques have also been widely applied to predict cement compressive strength using chemical composition and physical property data. Studies by Ramadhan and Raharjo showed that machine learning models trained on cement chemical and physical parameters can accurately predict ordinary Portland cement compressive strength [15]. Similarly, Olubajo and Paul applied Response Surface Methodology to optimize cement blends incorporating supplementary materials, improving mechanical strength performance [16]. However, many existing studies rely mainly on laboratory test data and do not incorporate environmental conditions or real-time production parameters. Recent research highlights the importance of integrating process parameters, environmental factors, and surface image analysis to improve prediction reliability and support multi-age strength prediction models for industrial applications [24].

Recent developments in deep learning have also enabled automated classification of cement clinker phases with improved accuracy and consistency. The composition and distribution of clinker phases strongly influence cement hydration behavior and mechanical performance. Deep learning–based image classification models have demonstrated strong capability in identifying primary clinker phases directly from microscopic images, providing a reliable alternative to manual microscopic analysis [14], [18]. These automated approaches improve repeatability, reduce analysis time, and support real-time industrial quality monitoring systems.

Despite these advancements, most existing research focuses on solving individual problems such as crack detection, strength prediction, or phase classification separately. There is still a lack of integrated systems that combine image-based material analysis, IoT-based measurement, and machine learning–based prediction into a unified platform for real-time cement quality monitoring and decision support. This research aims to address this gap by developing an integrated AI- and IoT-based framework for automated cement quality assessment.

III. Methodology

A. Color-Based Particle Identification

The proposed color-based particle identification system integrates a deep learning model, a backend service, and a web-based user interface. A YOLOv9-based instance segmentation model is used to detect and classify cement raw meal particles into three predefined color categories: dark red, light red, and white, representing different material characteristics.

The dataset consists of approximately 300 microscopic images collected from the quality control laboratory of INSEE Cement (Private) Limited, Sri Lanka. Images were manually annotated according to the defined color categories. Prior to training, preprocessing steps such as image resizing and pixel value normalization were applied to ensure consistent input dimensions. A FastAPI-based RESTful backend performs real-time inference on uploaded images, while a React-based frontend enables users to upload images, visualize detected particles, and store analysis results.

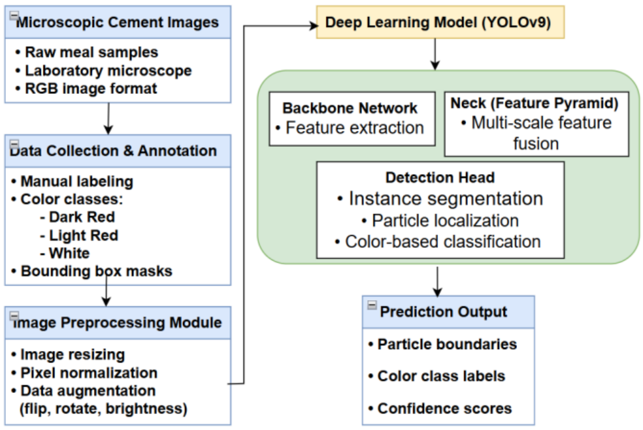

Fig. 1. Model Architecture of Color-Based Particle Identification

B. IoT-Based Area Measurement and Image-Based Crack Detection

The system incorporates non-contact ultrasonic sensors to measure the length and width of cement cube faces, following established sensor performance studies for distance and surface measurement applications [22]. The sensor readings are processed in real-time, and the calculated area of the cube face is stored automatically. A camera module is used to continuously monitor the cube during compressive loading. The images captured are processed through computer vision techniques using OpenCV and classified using a Convolutional Neural Network (CNN). The system integrates a load cell with an HX711 amplifier to capture the applied compressive force. By combining force readings with the measured surface area, the system automatically calculates compressive strength using the formula:

$$\sigma = F / A. \tag{1}$$

where σ represents compressive strength (MPa), F is the applied load (N), and A is the cross-sectional area (m²).

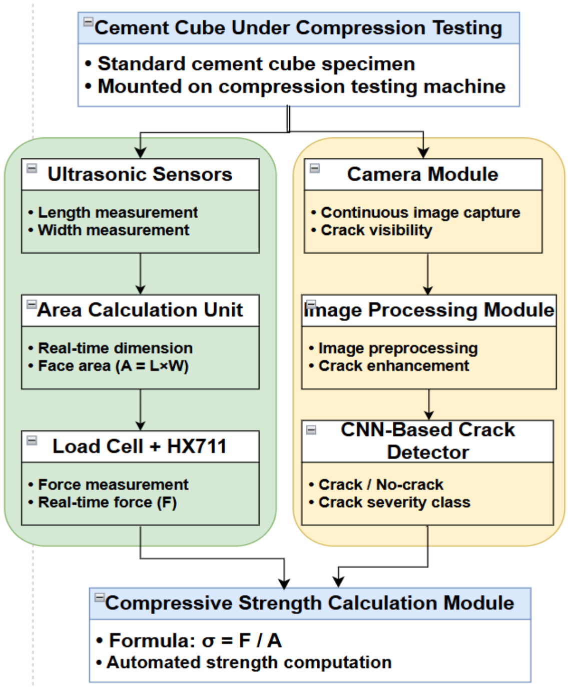

Fig. 2. Model Architecture of Area Measurement and Image-Based Crack Detection

C. Cement Strength Predictor with Environmental Sensitivity

This study utilizes industrial OPC production data collected from Sri Lankan cement plants, including chemical composition, physical properties, curing conditions, and compressive strength measurements at 1, 2, 7, 28, and 56 days. The dataset was cleaned, normalized, and enhanced through feature engineering, where clinker mineral phases were calculated using Bogue equations to better represent cement hydration behavior. Environmental factors such as temperature and humidity were incorporated to improve prediction reliability under real production conditions.

A hybrid ensemble machine learning framework combining XGBoost and LightGBM was developed using a multi-output regression approach to predict compressive strength at multiple curing ages simultaneously. XGBoost captured complex nonlinear interactions, while LightGBM ensured efficient learning and generalization. The model's performance was evaluated using standard regression metrics, demonstrating its effectiveness for accurate strength prediction and real-time quality control applications. The dataset was split into training (80%) and testing (20%) sets, and model performance was evaluated using cross-validation to ensure robustness.

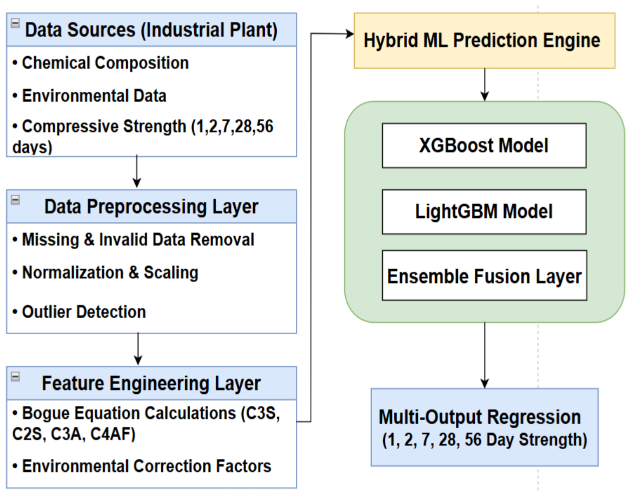

Fig. 3. Model Architecture of Cement Strength Predictor

D. Automated Classification of Cement Clinker Phases

The system utilizes a convolutional neural network (CNN) with transfer learning based on MobileNetV2 architecture to classify four primary clinker phases: Dicalcium Silicate (C_2S), Tricalcium Silicate (C_3S), Tricalcium Aluminate (C_3A), and Tetracalcium Aluminoferrite (C_4AF). A FastAPI backend is developed to provide real-time predictions, while a React-based frontend offers an interactive user interface for visualization and analysis.

Data Collection and Preprocessing: Microscopic images of cement clinker samples were collected from the laboratory of INSEE Cement (Private) Limited in Sri Lanka. The dataset consisted of approximately 500 images. Preprocessing included image resizing and normalization.

Model Training: MobileNetV2 was selected for its efficiency in image classification tasks. The model was trained using transfer learning.

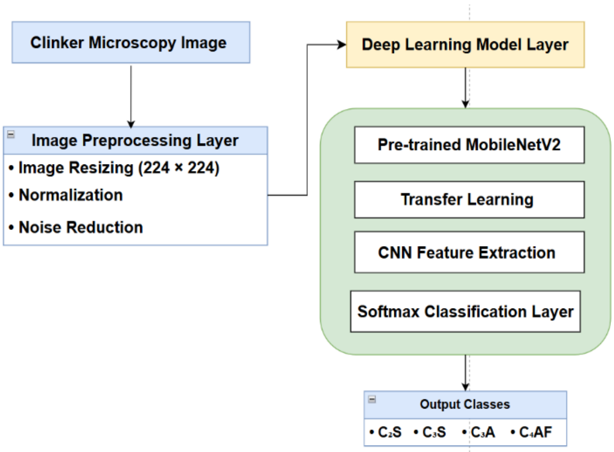

Fig. 4. Model Architecture of Cement Clinker Phases Classification

IV. Results and Discussion

A. Color-Based Particle Identification in Clinker Microscopy

The final model, named YOLOv9 with Advanced Object Detection, combines state-of-the-art YOLO architecture with optimized hyperparameters to enhance particle detection accuracy. This approach achieved a mean Average Precision (mAP@0.5) of 86.45% as shown in TABLE I in detecting and classifying cement raw meal particles, including dark red, light red, and white particle categories.

TABLE I. ACCURACY WITH EXPERIMENTED MODELS

Model	Train Accuracy	Validation Accuracy
YOLOv5s	78.45%	75.80%
YOLOv8n	82.30%	77.47 %
YOLOv8s	84.15%	81.20%
YOLOv9c	88.92%	86.45%
YOLOv9e	91.25%	88.30%

Fig. 5 demonstrates the accuracy graph, highlighting the improvement in mean Average Precision over the training epochs, while Fig. 6 depicts the loss graph, showcasing how effectively the model learns and improves over time.

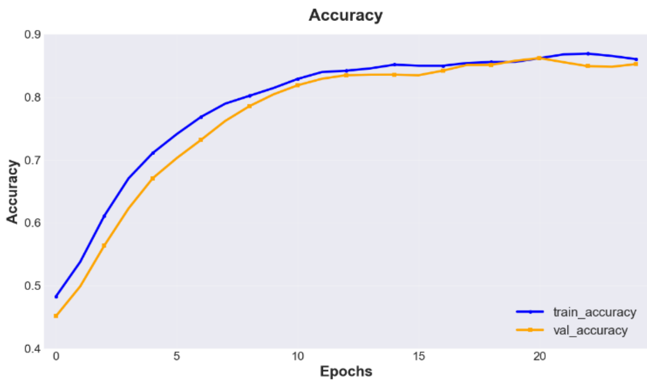

Fig. 5. Training and validation Accuracy Graph of Color-Based Particle Identification

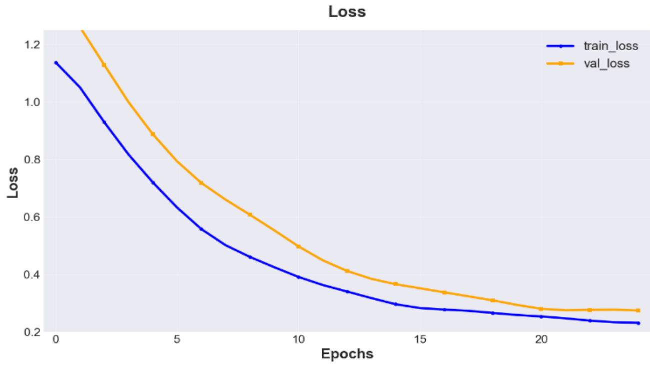

Fig. 6. Training and validation loss of Color-Based Particle Identification

These visualizations confirm that the optimized YOLOv9 architecture effectively learns discriminative particle features, resulting in accurate color-based particle identification in clinker microscopy images.

B. IoT-Based Area Measurement and Crack Detection for Cement Cube Testing

The final model, utilizing U-Net deep learning architecture with optimized encoder-decoder structure, was developed for automated crack detection and segmentation in cement cube specimens. This approach integrates semantic segmentation with pixel-level classification to accurately identify and quantify crack patterns in concrete surface images. The optimized U-Net model achieved exceptional performance with an accuracy of 92.35% on the training set and 90.15% on the validation set, as shown in TABLE II.

TABLE II. ACCURACY WITH EXPERIMENTED MODELS

Model	Train Accuracy	Validation Accuracy
U-Net Basic	87.25 %	85.10%
U-Net++	89.40 %	86.75%
DeepLabV3	85.80 %	83.20%
SegNet	84.50 %	82.90%
U-Net Optimized	92.35 %	90.15 %

The U-Net Optimized architecture demonstrates superior performance compared to other segmentation models, with a 5-8% improvement over the basic U-Net implementation and 6-10% improvement over DeepLabV3 and SegNet architectures. The model's encoder-decoder structure with skip connections enables precise crack boundary detection at the pixel level. Data augmentation techniques such as rotation and flipping were applied to improve model generalization and reduce overfitting.

Fig. 7 demonstrates the accuracy graph, highlighting the improvement in segmentation accuracy over the training epochs, while Fig. 8 depicts the loss graph, showcasing how effectively the model learns and improves over time.

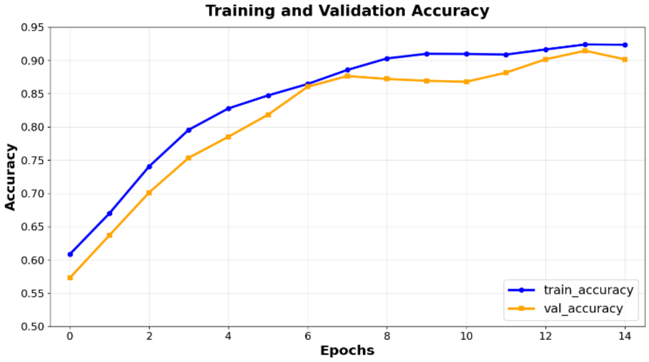

Fig. 7. Training and validation Accuracy Graph of Crack Detection

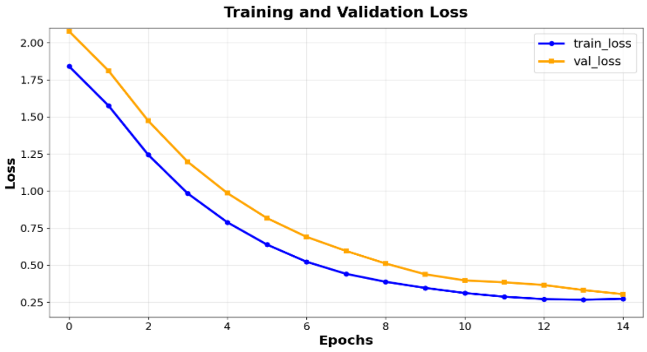

Fig. 8. Training and validation loss of Crack Detection

These visualizations confirm that the U-Net Optimized architecture with advanced skip connections and batch normalization resulted in a highly effective model for automated crack detection and structural integrity assessment in cement cube testing.

C. Cement Strength Prediction (Multi-Output)

The final ensemble model, combining XGBoost and LightGBM algorithms, was developed for multi-output cement strength prediction at five critical time intervals (1D, 2D, 7D, 28D, and 56D). This approach incorporates 14 input parameters including physical properties (initial grinding time, final grinding time, residue 45µm, and fineness) and chemical composition with over 40 engineered features including interaction terms, ratio features, polynomial transformations, and logarithmic transformations. The ensemble model achieved an average validation R^2 score of 0.8896, indicating strong predictive capability across all curing ages.

TABLE III. MODEL PERFORMANCE COMPARISON

Model	Train R^2 Score	Validation R^2 Score
XGBoost Only	90.14%	85.66%
LightGBM Only	89.05%	86.20%
Random Forest	85.30%	87.50%
Gradient Boosting	87.60%	79.87%
Ensemble (XGBoost + LightGBM)	91.48%	88.96%

The advanced feature engineering strategy played a crucial role in improving model performance. Over 40 derived features were generated, including interaction terms (e.g., $CaO \times SiO_2$), ratio features (e.g., CaO/SiO_2), aggregate oxide indicators, polynomial transformations, cube-root scaling, and logarithmic transformations.

The final model's accuracy and loss curves are presented in Fig. 9 and Fig. 10. respectively. The hybrid model maintained consistent performance across both training and validation sets, indicating effective generalization to unseen data.

Fig. 9. Model Performance Curve (R^2 Score)

Fig. 10. Training and validation loss of Strength Predictor

D. Automated Classification of Cement Clinker Phases

The final model, utilizing MobileNetV2 with Transfer Learning architecture, was developed for automated classification of cement clinker phases in microscopic images. This approach integrates pre-trained ImageNet weights with domain-specific fine-tuning to enhance feature extraction and classification accuracy. The model successfully classifies four critical cement phases: C2S (Belite - Dicalcium Silicate), C3A (Tricalcium Aluminate), C3S (Alite - Tricalcium Silicate), and C4AF (Brownmillerite - Tetracalcium Aluminoferrite), achieving an accuracy of 93.85% on training and 91.60% on validation as shown in TABLE IV.

TABLE IV. ACCURACY WITH EXPERIMENTED MODELS

Model	Train Accuracy	Validation Accuracy
VGG16	82.45%	80.20%
ResNet50	85.30%	83.15%
InceptionV3	88.75%	86.40%
MobileNetV2	89.50%	87.80%
MobileNetV2 + Transfer Learning	93.85%	91.60%

The MobileNetV2 with Transfer Learning architecture demonstrates superior performance compared to other deep learning models, with an 11% improvement over VGG16 and 5-8% improvement over InceptionV3 and standard MobileNetV2. The model's lightweight architecture combined with transfer learning enables efficient feature extraction while maintaining high classification accuracy across all four cement clinker phases.

Fig. 11 demonstrates the accuracy graph, highlighting the improvement in classification accuracy over the training epochs, while Fig. 12 depicts the loss graph, showcasing how effectively the model learns and improves over time.

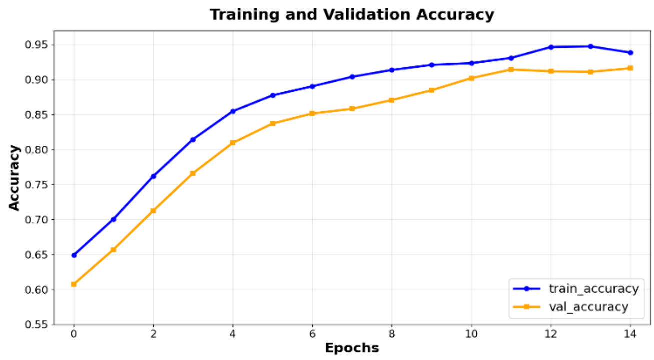

Fig. 11. Training and validation Accuracy Graph of Classification of Cement Clinker Phases

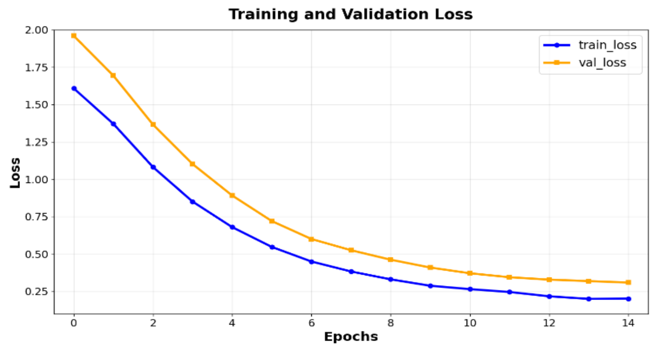

Fig. 12. Training and validation loss of Classification of Cement Clinker Phases

V. CONCLUSION AND FUTURE WORK

This research proposes an integrated AI- and IoT-based framework for automated cement quality assessment, combining particle identification, crack detection, clinker phase classification, and multi-output cement strength prediction within a unified system. Experiments using real industrial data demonstrated strong predictive performance, faster analysis, and improved reliability compared to traditional manual testing methods. The proposed framework is designed to be adaptable to different cement manufacturing environments by incorporating scalable machine learning models and configurable IoT components. Although validated using data from a single industrial source, the system can be extended to other plants with minimal retraining using locally collected data. Future work will focus on supporting additional cement types and supplementary materials, integrating real-time production sensor data, and deploying the system as a scalable edge-enabled solution for continuous industrial monitoring.

REFERENCES

[1] U.S. Geological Survey, "Mineral Commodity Summaries: Cement," U.S. Department of the Interior, 2024.

[2] P. Kumar and R. Kumar, "Quality control and assurance in cement manufacturing," Construction and Building Materials, vol. 185, pp. 10–18, 2018.

[3] H. F. W. Taylor, Cement Chemistry, 2nd ed. London, U.K.: Thomas Telford Publishing, 1997.

[4] P. Hewlett and M. Liska, Lea's Chemistry of Cement and Concrete, 5th ed. Oxford, U.K.: Butterworth-Heinemann, 2019.

[5] A. M. Neville, Properties of Concrete, 5th ed. London, U.K.: Pearson Education, 2011.

[6] Y. Yeh, "Modeling of strength of high-performance concrete using artificial neural networks," Cement and Concrete Research, vol. 28, no. 12, pp. 1797–1808, 1998.

[7] Z. Zhang, S. Y. Lee, and J. Park, "Deep learning-based image analysis for concrete crack detection," Automation in Construction, vol. 112, pp. 103–114, 2020.

[8] S. C. Mukhopadhyay and N. K. Suryadevara, "Internet of Things: Challenges and opportunities," IEEE Potentials, vol. 33, no. 3, pp. 35–40, 2014.

[9] A. A. Ramezanianpour, M. Shekarchi, and M. Khayat, "Quality control challenges in cement and concrete industries," Construction and Building Materials, vol. 25, no. 1, pp. 345–354, 2011.

[10] K. Singh et al., "Machine learning applications in cement quality control: A comprehensive review," Cement and Concrete Research, vol. 142, pp. 106-118, 2023.

[11] L. Chen et al., "Crack detection in concrete structures using deep learning," Sustainability, vol. 14, no. 13, pp. 8117-8132, 2022.

[12] M. Wang and R. Kumar, "Machine learning in concrete science: applications," Journal of Materials Science, vol. 58, pp. 45-60, 2023.

[13] S. Anderson et al., "Machine learning in concrete technology: A review of current researches, trends, and applications," Frontiers in Built Environment, vol. 9, pp. 1145591, 2023.

[14] T. Wilson and P. Davis, "Automated analysis of cement clinker microscopy images using computer vision," Cement and Concrete Composites, vol. 125, pp. 104, 2024.

[15] S. Ramadhan and A. B. Raharjo, "Prediction of OPC Cement Compressive Strength Based on Cement Chemical and Physical Parameters Using Machine Learning Techniques," JRSSEM, vol. 4, no. 12, 2025.

[16] O. O. Olubajo and A. S. Paul, "Strength Prediction and Optimisation of Velvet Tamarind Pod Ash Cement Blends via Response Surface Methodology," Abubakar Tafawa Balewa University, 2024.

[17] B. Yang et al., "Analysis and prediction of compressive strength of calcium aluminate cement paste based on machine learning," Archives of Civil and Mechanical Engineering, vol. 25, no. 1, 2024.

[18] T. Wilson and P. Davis, "Automated analysis of cement clinker microscopy images using computer vision," Cement and Concrete Composites, vol. 125, pp. 104-115, 2024.

[19] U. Patel and V. Sharma, "Color-based particle identification in cement clinker using machine learning," Construction and Building Materials, vol. 389, pp. 131-142, 2023.

[20] H. K. Kapadia, P. V. Patel, and J. B. Patel, "Convolutional neural network based improved crack detection in concrete cubes," International Journal of Computing and Digital Systems, vol. 13, no. 1, pp. 341-352, 2023.

[21] M. S. Patil, R. B. Ghongade, and R. V. Salunke, "Image processing and Machine learning in Concrete Cube Crack detection," in 2023 International Conference on New Frontiers in Communication, Automation, Management and Security (ICCAMS), vol. 1, pp. 1-5, 2023.

[22] A. M. Kassim, H. I. Jaafar, M. A. Azam, N. Abas, and T. Yasuno, "Performances study of distance measurement sensor with different object materials and properties," in 2013 IEEE 3rd International Conference on System Engineering and Technology, pp. 281-284, 2013.

[23] M. Gkantou, M. Muradov, G. S. Kamaris, K. Hashim, W. Atherton, and P. Kot, "Novel electromagnetic sensors embedded in reinforced concrete beams for crack detection," Sensors, vol. 19, no. 23, pp. 5175, 2019.

[24] G. Doğan, A. Özkiş, and M. H. Arslan, "A new methodology based on artificial intelligence for estimating the compressive strength of concrete from surface images," Ingeniería e Investigación, vol. 44, no. 1, pp. 4, 2024.